

Using Computer Vision to Translate Sign Language into Words

Yukshing Pang, Shanghai Starriver Bilingual School

Abstract

Computer vision is a subdomain of artificial intelligence focusing on making computers perceive and understand different visual scenes. It encompasses extracting, analyzing, and understanding meaningful information from digital images or videos. After combining it with the neural network, it can dispose of vast information and perform well in image classification, object detection, scene flow estimation, and other significant fields. Sign language translation or recognition (SLR) is a rarely-noticed task that computer vision can also do. Sign languages are visual languages that convey information by signers' hands, facial expressions, body movements, and so forth. Differentiating a series of different actions and expressions is a difficult video classification task. This research focused on using computer vision to recognize World-Level American Sign Language (WLASL) signs from videos and translate them into English words. Moreover, this research will use skeleton-based SLR to solve the task based on previous research.

Keywords: Sign language recognition (SLR), Sign language translation, World Level American Sign language dataset, Computer vision, Neural network

Introduction

About 5% of the world's population is reported to be deaf and mute, and approximately eight people in every 10,000 are born mute. Sign language has become one of the main methods for them to communicate with others; however, many of us do not understand sign language. Due to this phenomenon, a translator that can translate sign language into words is needed. It can help us to understand and look after muted



Figure 1. WLASL

people better, help people to understand deaf and speech-impaired people, and make their lives easier.

Sign language is a visual language performed with the dynamic movement of hand gestures, body posture, and facial expressions. Because of its complications, it is hard to establish a well-performing model that can recognize sign language with high accuracy. The traditional SLR has a broad range of applications including sign spotting, sign video retrieval, sign language translation, and continuous sign language recognition; however, the accuracy of traditional SLR (3D-CNN, RNN, etc.) is not high enough for application in the commercial field.

Recently, skeleton-based action recognition has become popular and its core idea matches the requirement of SLR. Based on previous studies, the precision of SLR would increase obviously if they use skeleton-based action recognition in the model and some concepts of skeleton-based SLR come up.

This research chooses the best-performed skeleton-based SLR framework on the WLASL-2000 benchmark and compares it with other SLR models.

Related Works

Sign language recognition (SLR) classifies the isolated signs from videos into a set of glosses, trains these glosses, and identifies different signs. This is the basic task in the field of sign language understanding. However, a persistent challenge in SLR is the simultaneous capture of global body motion information and local details such as arm, hand, and facial expressions. This problem was solved both by recent research about natural language-assisted sign language recognition (NLA-SLR) and skeleton-based sign language recognition.

The natural language-assisted sign language recognition (NLA-SLR) framework mainly consists of three parts: 1) data pre-processing which generates video key point pairs as network inputs; 2) a video-key point network (VK Net) which takes video key point pairs of various temporal receptive fields as inputs for vision feature extraction; 3) a head network containing a language-aware label smoothing branch and an inter-modality mix-up branch which can be applied on both RGB videos and key point heatmap sequences.

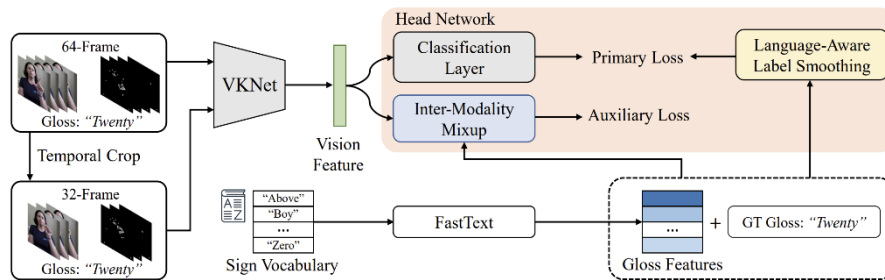


Figure 2, natural language-assisted sign language recognition (NLA-SLR)

Skeleton-based sign language recognition involves identifying unique patterns in human joint positions and movements. It works well on sign language recognition based on not only considering the main body coordinates but also providing ground truth annotations for hands.

In previous studies, scholars developed Sign Language Graph Convolution Network (SL-GCN), which is used to analyze the spatiotemporal skeleton graph, and Separable Spatial Temporal Convolution Network (SSTCN) which focuses on fully exploiting the information in whole-body key points and whole-body skeleton features. They use these models and form the natural language-assisted sign language Recognition (SAM-SLR) framework, which is the top framework in WLASL-2000 2022 and second in 2023.

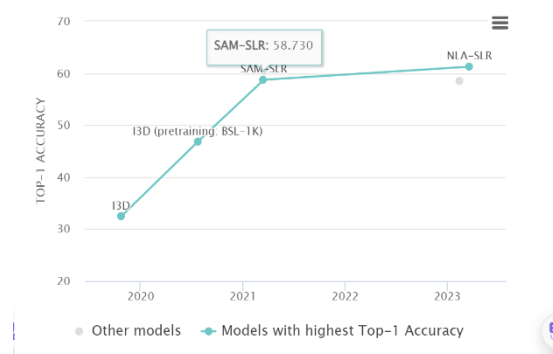


Figure 3, WLASL-2000 benchmark

World-level American sign language (WLASL) is a dataset that contains 2000 more common different words in sign language with clear videos and actions. This research will use this dataset to test and train the framework.

Method

SL-GCN model will extract motion dynamics from the graph by attention mechanism. The processes are 1. Input sign language videos 2. SLR graph is constructed from whole-body key points after graph reduction 3. Trained by the multi-stream SL-GCN model. In SL-GCN, a pre-trained whole-body pose estimation network provides 133 key points estimated from the detected person in videos. The number of key points is decreased to 27 because a large number of key points will produce noise and decrease the accuracy of the model.

Multi-stream arithmetic is also developed and used in the model. Key points are the joints of the human body and bones are the distance between key points. Multi-stream SL-GCN analyzes joints, bones, joint motions, and bone motions in different workflows. Every key point in the video frame has a coordinate position. For the whole video frame of the human body, the key point of the nose is identified as the center which can be represented as $v = (0,0,0)$. Other targets can be represented as $v = (x, y, z)$ (z represents confidence score). The bone vectors of the other nodes can be calculated by subtracting their source joint coordinates from their current joint coordinates. Motion data are generated by calculating the difference between adjacent frames in both joint and bone streams.

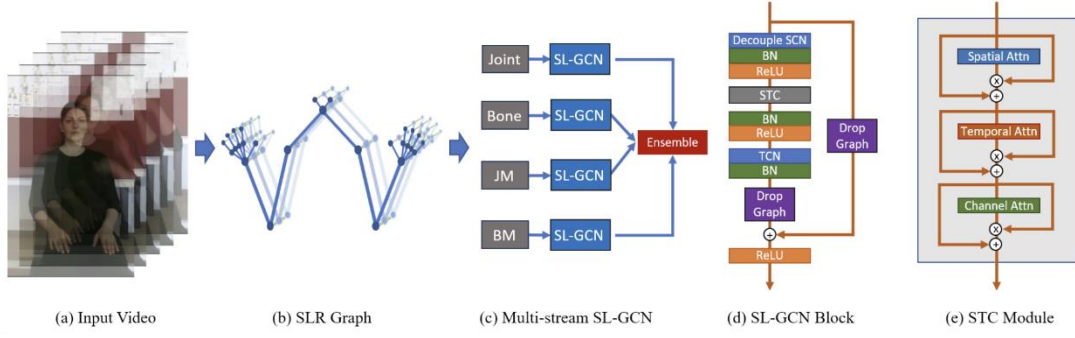


Figure 3, SL-GCN

Every SL-GCN block consists of a decoupled spatial convolutional layer (Decouple SCN), an STC (spatial, temporal, and channel-wise) attention module, a temporal convolutional layer (TCN), and a Drop-Graph module. These layers can avoid over-fitting and decrease the calculated amount. Every STC module consists of a spatial attention layer, a temporal attention layer, and a channel attention layer. The whole SL-GCN model has 10 basic blocks, global average pooling layers, and a fully connected layer.

SSTCN model recognizes sign language videos by features. This can help analyze the features displayed in different frames of the video and focus on the changes that happened on these features but not the relationship between them, as SL-GCN has done. 33 key points are added into 60 frames of the video. With 1 landmark on the nose, 4 landmarks on the mouth, 2 landmarks on the shoulders, 2 landmarks on elbows, 2 landmarks on the wrists, and 22 landmarks on the hands, the video is inputted into the model. There are four stages in the model. Stage 1 reshapes the

features from $60 \times 33 \times 24 \times 24$ to $60 \times 792 \times 24$ and put them into two layers of 1×1 kernels to gain temporal information. In Stage 2 and Stage 3, features are inputted into two layers of 3×3 kernels. At stage 4, a couple of 3×3 fully connected layers are implemented to generate prediction features.

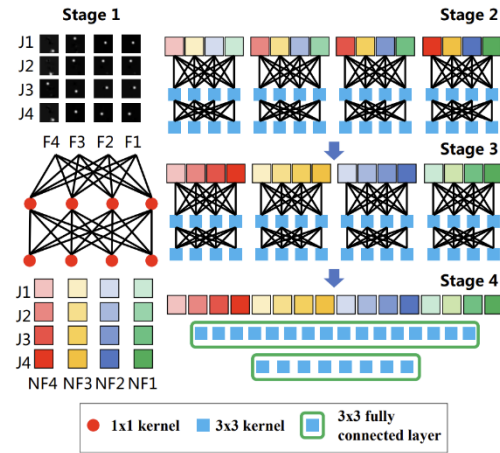


Figure 4, SSTCN

Result

After combining these models with a simple ensemble, a weight is added and used to train the model. As confirmed, the SL-GCN model is effective. The combination of all streams increases the overall accuracy and reaches the level of

previous research. The SSTCN model also performs well. Compared to

Streams	Top-1	Top-5
Joint	95.02	99.21
Bone	94.70	99.14
Joint Motion	93.01	98.85
Bone Motion	92.49	98.78
Multi-stream	95.45	99.25

Figure 5, SL-GCN all streams' result

Methods	Feature size	Top-1
ResNet3D	12×12	92.82
ResNet2+1D	12×12	93.03
SSTCN	12×12	93.60
SSTCN	24×24	94.32

Figure 6, SSTCN's result

ResNet3D and ResNet2+1D, the accuracy of SSTCN is slightly better. Moreover, the SSTCN model performs better if the feature size increases as shown in the table. These results match the result of the pre-trained model provided by the research group.

Discussion

In the process of reproducing the two models, although there are pre-trained models provided, there are also several difficulties and obstacles. The Github code provided by the previous research group is not simple and clear. Readers have to be experienced scholars to comprehend the structures and files directly. A huge amount of time was spent on solving errors and debugging when I managed the code. For instance, when I was trying to clone the Github

code into a Tencent cloud server that I borrowed from the platform, I tried both https and ssh methods but errors always came out and no solution on the internet was effective in solving this problem. To solve the problem, I downloaded the Zip file of the code and uploaded it to the cloud server at last. In conclusion, I spent an unbroken day only solving the problem of cloning code into the cloud server which sound very easy. The purpose of talking about this experience is to avoid new people facing the same problem, wasting a great amount of time to solve this, and spending their time on solving significant problems in their research.

Back to the research topic, skeleton-based SLR has great potential for translating sign language. Although the overall accuracy of SAM-SLR in WLASL-2000 is not enough, this accuracy can improve in the future when a better skeleton-based action recognition framework is invented or a better skeleton graph is created. This technology has great potential to improve people's quality of life and commercial development.

Conclusion

In this paper, I reproduce two models of skeleton-based SLR, which are SL-GCN and SSTCN in SAM-SLR, proving their rationality and comparing them to other models. Both models work well and perform best in their field. In the future,

these two models can be improved by improving the structure or lost functions and creating a better skeleton graph. There are also some drawbacks to these models. At first, all of these models only translate sign language into words, but our purpose is to translate these sign languages into sentences with clear grammar. Translating words is only the first step of this project. Secondly, videos in WLASL and other datasets are high-quality. There are lots of interference factors in real life and anti-interference performance needs to improve.

References

- Skeleton Aware Multi-modal Sign Language Recognition, Songyao Jiang, Bin Sun , Lichen Wang, Yue Bai, Kunpeng Li and Yun Fu Northeastern University, Boston MA, USA
- Natural Language-Assisted Sign Language Recognition, Ronglai Zuo¹ Fangyun Wei Brian Mak¹
¹The Hong Kong University of Science and Technology
²Microsoft Research Asia
- WLASL-2000 Benchmark (Sign Language Recognition) | Papers With Code
- <https://github.com/jackyjsy/CVPR21Chal-SLR>
- Jinmiao Cai, Nianjuan Jiang, Xiaoguang Han, Kui Jia, and Jiangbo Lu. Jolo-gcn: Mining joint-centered light-weight information for skeleton-based action recognition. In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, pages 2735–2744, 2021.
- Yong Du, Wei Wang, and Liang Wang. Hierarchical recurrent neural network for skeleton-based action recognition. In Proceedings of the IEEE conference on computer vision and pattern recognition, pages 1110–1118, 2015.
- Prajit Ramachandran, Barret Zoph, and Quoc V Le. Searching for activation functions. arXiv preprint arXiv:1710.05941, 2017.
- Lei Shi, Yifan Zhang, Jian Cheng, and Hanqing Lu. Skeleton-based action recognition with multi-stream adaptive graph convolutional networks. IEEE Transactions on Image Processing, 29:9532–9545, 2020
- Yi-Fan Song, Zhang Zhang, Caifeng Shan, and Liang Wang. Stronger, faster, and more explainable: A graph convolutional baseline for skeleton-based action recognition. In Proceedings of ACM International Conference on Multimedia, pages 1625–1633, 2020.

- Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. Dropout: a simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(1):1929–1958, 2014